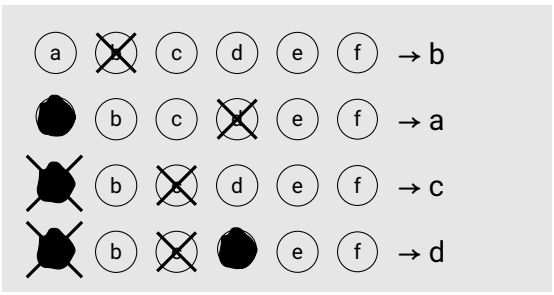
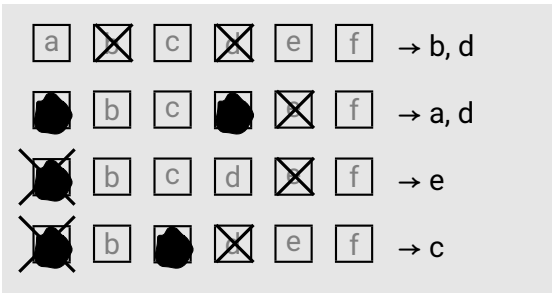
Exercises

1	2	3	4	5	6	7	8

Surname, First name

Machine Learning 1 (52041MAL6Y)
Hertentamen (moved from 07-01-2026 to 14-01-2026)
7 Jan, 2026, 14:00-17:00





Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Pay attention to the following;

- **WRITE YOUR NAME AND STUDENT NUMBER ON THE FRONT PAGE** (don't forget to mark the digits)
- Write all your answers on this exam booklet; it will be scanned and digitally graded.
- Write your **answers inside the boxes** (it is ok, to slightly go over the margins though).
- The answer boxes should be large enough for your answer
- **If you need to empty the answer box in order to start over, ask the invigilator for a blank sticker or use the *Emergency Box*at the end of the booklet.**

During the exam you are allowed to use:

- The provided cheat sheet.
- Basic calculator (although you will not need it)

Please scan over the questions before you start to get an impression of the content

- You have 3 **hours** for the exam (except for those with pre-approved extensions)
- In total **40 points** can be earned divided over 7 exercises:
  - Exercise 1: Multiple choice questions [10p]
  - Exercise 2: True/False statements [6p]
  - Exercise 3: MAP estimation at Dunder Mifflin Paper Co. [4p]
  - Exercise 4: Regularized Ridge Regression [4p]
  - Exercise 5: 1D classification with SVMs [4p]
  - Exercise 6: Classification and dimensionality reduction on the MNIST Dataset [4p]
  - Exercise 7: Mixture of Experts for modelling customer behaviour [8p]
- We work with a **partial grading**: you can get points even if you don't manage to solve the full question. Hint clearly define your (intended) steps, even when you get stuck in a derivation.

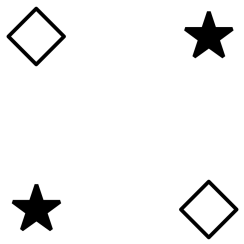
[10p] Multiple choice questions

1.5p **1a** Consider a D-dimensional dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N\}, \mathbf{x}_n \in \mathbb{R}^D$, with empirical covariance matrix $\mathbf{S}$. Let $\mathbf{U}_M$ be a matrix whose columns are the eigenvectors corresponding to the $M < D$ largest eigenvalues of $\mathbf{S}$. Now, consider the dataset $\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, ..., \mathbf{z}_N\}$ where $\mathbf{z}_n = \mathbf{U}_M^T(\mathbf{x}_n - \bar{\mathbf{x}})$, where $\bar{\mathbf{x}}$ is the sample mean of $\mathbf{X}$.

Which of the following statements about dataset $\mathbf{Z}$ are true?

- ☐ It has zero mean.
- ☐ Its covariance matrix is the identity matrix.
- ☐ It has the same dimensionality as $\mathbf{X}$.
- ☐ Its features are decorrelated.

1.5p **1b** Which classifiers can achieve zero training error on the following labeled 2D dataset?



Note: Unless stated otherwise, we use the original features as input (i.e., without applying basis functions).

- ☐ Perceptron.
- ☐ Multi-layer Perceptron with 1 arbitrarily large hidden layer and *linear activation function*.
- ☐ Linear Discriminant Analysis.
- ☐ Quadratic Discriminant Analysis.
- ☐ Logistic Regression.
- ☐ Support Vector Machine with quadratic kernel.

1.5p **1c** Which of the following statements regarding Lasso and Ridge regression are true?

- ☐ In both Lasso and Ridge regression, a penalty term is added to the loss function to reduce model variance and help prevent overfitting.
- ☐ The penalty term in Ridge regression can be interpreted as imposing a Gaussian prior on the coefficients.
- ☐ The penalty term in Lasso regression can be interpreted as imposing a Gaussian prior on the coefficients.
- ☐ Ridge regression can be used for feature selection because it can shrink some coefficients exactly to zero.
- ☐ Lasso regression can be used for feature selection because it can shrink some coefficients exactly to zero.

We consider the following soft-margin SVM optimization problem (relevant for questions 1d and 1e):

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{n=1}^{N}\xi_n \text{ s.t } \begin{cases} t_n\left(\mathbf{w^T x_n} + b\right) & \geq 1-\xi_n \quad \forall n \in \{1,2,...,N\}, \\ \xi_n & \geq 0 \quad \forall n \in \{1,2,...,N\}. \end{cases}$$

1.5p **1d** If we set the hyperparameter C to a very large value ($C \longrightarrow \infty$), which of the following will occur?

- ☐ The number of support vectors will increase.
- ☐ The number of support vectors will decrease.
- ☐ The model will behave as a hard-margin SVM.
- ☐ The model will strictly prevent overfitting.
- ☐ All optimal slack variables ξ_n^* will be equal to 1.

1p **1e** What does it mean if a specific slack variable $\xi_n^* > 1$?

- (a) The data point $\mathbf{x}_n$ is correctly classified and lies outside the margin.
- (b) The data point $\mathbf{x}_n$ is correctly classified but lies inside the margin.
- (c) The data point $\mathbf{x}_n$ lies exactly on the decision boundary.
- (d) The data point $\mathbf{x}_n$ is misclassified.

1p **1f** Consider a continuous uniform distribution $U[0, b]$, from which we draw a random real number between 0 and b. We sample three points form $U[0, b]$, obtaining $x_1 = 30.1$, $x_2 = 8.7$ and $x_3 = 21.2$. Based on those three samples, what is the maximum likelihood estimate of b?

(a) $\hat{b}_{ML} = 60.$

(b) $\hat{b}_{ML} = 20.$

(c) $\hat{b}_{ML} = 30.1.$

(d) $\hat{b}_{ML} = 21.2.$

(e) $\hat{b}_{ML} = 8.7.$

1p **1g** Consider a perfectly balanced binary classification dataset with N samples (i.e. there are exactly $\frac{N}{2}$ samples of each class) and a probabilistic model that always predicts a probability of 0.5 for both classes, no matter what the input is. What is the cross-entropy loss for this model and dataset?

(a) $0.$

(b) $\frac{N}{2}.$

(c) $N \log(2).$

(d) $N.$

1p **1h** Consider a Gaussian Process $f(\cdot) \sim GP(m(\cdot), k(\cdot, \cdot))$ defined by a mean function $m(x) = 0$ and an exponential (RBF) kernel:

$$k(x_n, x_m) = \theta_0 \exp\left(-\frac{1}{2\theta_1^2} \|x_n - x_m\|^2\right)$$

Which of the following statements is **TRUE**?

(a) For any finite set of points $\{x_1, \ldots, x_N\}$, the joint distribution of the function values $p(f(x_1), \ldots, f(x_N))$ is a multivariate Gaussian distribution.

(b) Increasing the length-scale hyperparameter θ_1 results in sampled functions that look more "wiggly" and decorrelate more quickly over distance.

(c) The Gaussian Process is typically considered a parametric model because its behavior is determined by a fixed number of hyperparameters (θ_0, θ_1).

(d) The predictive variance σ' of $f(x')$ at a test point x' increases as we add more training observations in the immediate vicinity of x'.

[6p] True or False

Determine whether the following statements are True or False. **Justify your answer** by giving a proof/explanation or a counterexample. Note that you will only get points if your explanation is correct.

1.5p **2a** Let $\mathbf{x}_1, \ldots, \mathbf{x}_n \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be i.i.d. random vectors in $\mathbb{R}^d$. Consider the empirical mean $\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i$.

The estimator $\hat{\boldsymbol{\mu}}$ is an unbiased estimator of the true mean vector $\boldsymbol{\mu}$.

Circle your answer: True / False

1.5p **2b** Consider a mixture model $p(\boldsymbol{x_n}) = \sum_{k=1}^{K} \pi_k p(\boldsymbol{x_n}|\boldsymbol{\theta_k})$, where all the experts $p(\boldsymbol{x_n}|\boldsymbol{\theta_k})$ follow a probability distribution of the same type.

The number of parameters of the model increases linearly with respect to the number of experts (K).

Circle your answer: True / False

1.5p **2c** *If a Mixture of Gaussians can correctly partition a certian dataset according to its class assignments, i.e., consistent with ground-truth labels, K-means can also do so.*

Circle your answer: True / False

1.5p **2d** To properly estimate the generalization performance of a model, we should use the **test set** to tune hyperparameters (such as the regularization strength λ or the number of hidden layers).

Circle your answer: True / False

[4p] MAP estimation at Dunder Mifflin Paper Co.

Angela Martin (Head of Accounting) is tracking disruptions that affect office productivity to report them to corporate. She focuses on measuring the events that stop work across the branch.

Let X denote the number of times a major work interruption (e.g., an unauthorized fire drill, a Michael Scott mandatory party, or a prolonged Jim/Dwight prank) occurs during a business day. Assume $X \sim \text{Poisson}(\lambda)$, which has the probability density function

$$p(x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!},, \quad \lambda > 0, \quad x = 0, 1, 2, \ldots$$

Angela has a prior belief that λ is likely to be small, and models this belief with an Exponential prior ($\lambda \sim \text{Exponential}(\beta)$), which has density

$$p(\lambda \mid \beta) = \beta e^{-\beta\lambda}, \quad \lambda > 0,$$

where $\beta > 0$ is a known hyperparameter.
Angela wants to estimate the parameter λ based on a set of i.i.d. observations $\mathbf{x} = \{x_1, x_2, ..., x_N\}$.

2p **3a** Compute the log-posterior $\log p(\lambda \mid \mathbf{x})$ up to an additive constant.

2p **3b** Derive the Maximum a Posteriori (MAP) estimate of λ.

[4p] Regularized Ridge Regression

Consider a regression problem with training samples $\{\mathbf{x}_n\}_{n=1}^N$ and corresponding targets $\{t_n\}_{n=1}^N$. Let $\boldsymbol{\phi}$ be a vector of M basis functions, and define the design matrix

$$\boldsymbol{\Phi} = \begin{pmatrix} \boldsymbol{\phi}(\mathbf{x}_1)^\top \\ \boldsymbol{\phi}(\mathbf{x}_2)^\top \\ \vdots \\ \boldsymbol{\phi}(\mathbf{x}_N)^\top \end{pmatrix} \in \mathbb{R}^{N \times M},$$

and the target vector $\mathbf{t} = (t_1, t_2, \ldots, t_N)^\top$.

We fit a linear model

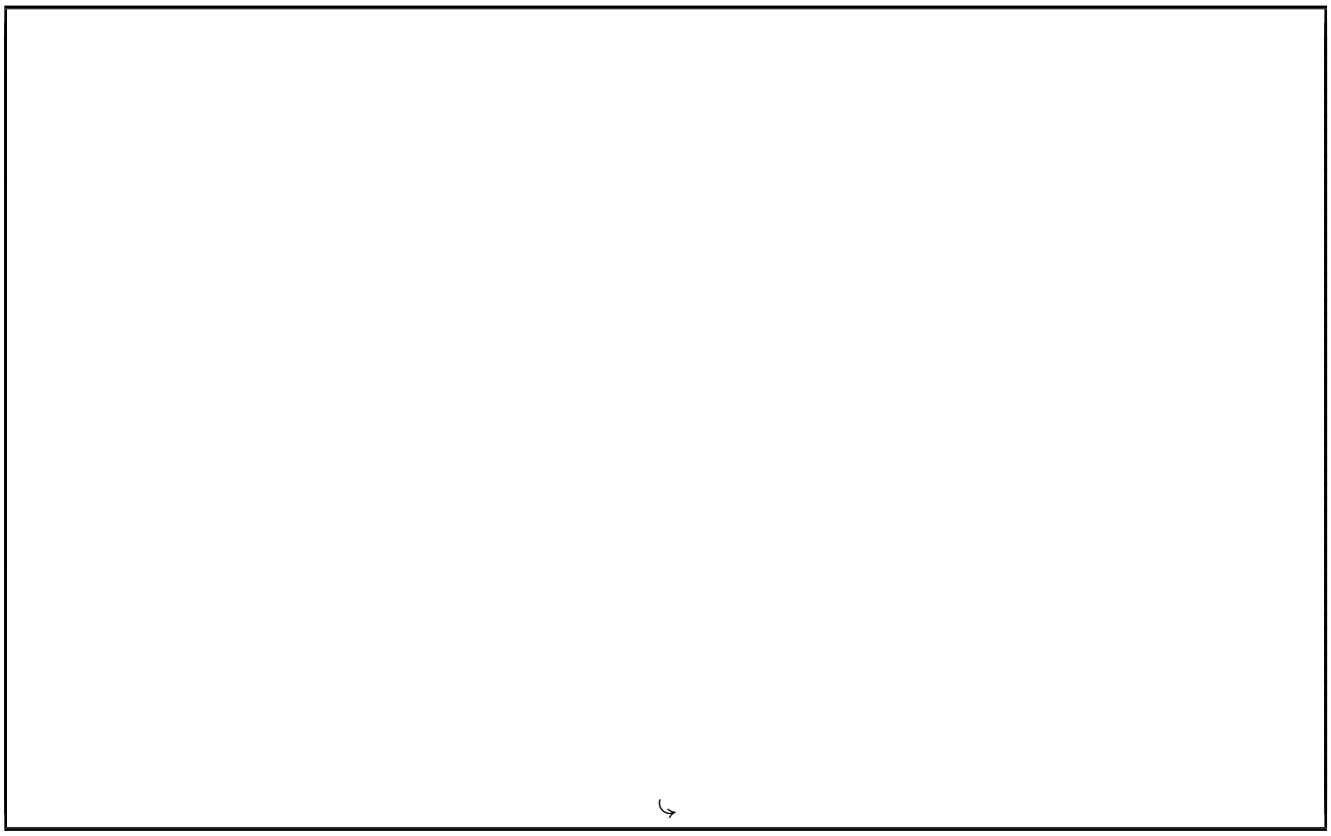
$$y(\mathbf{x}) = \mathbf{w}^\top \boldsymbol{\phi}(\mathbf{x})$$

by minimizing the regularized least squares objective

$$J(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \left(\mathbf{w}^\top \boldsymbol{\phi}(\mathbf{x}_n) - t_n \right)^2 + \frac{\lambda}{2} \mathbf{w}^\top \mathbf{w},$$

where $\lambda > 0$ is a regularization parameter.

3p **4a** Show that the closed-form solution for the weight vector is $\hat{\mathbf{w}} = \left(\boldsymbol{\Phi}\, \boldsymbol{\Phi}^\top + \lambda \mathbf{I} \right)^{-1} \boldsymbol{\Phi}^\top \mathbf{t}$
Note: a detailed solution with all steps is required for full points.



1p **4b** If we omit regularization $\lambda = 0$, which problem could we face when evaluating the closed-form solution for $\hat{\mathbf{w}}$?

[4p] 1D classification with SVMs

You are given 6 one-dimensional points $x_1 = -1$, $x_2 = 0$, $x_3 = 1$, $x_4 = -2$, $x_5 = 2$, $x_6 = 3$. Points x_1, x_2 and x_3 belong to class A, and x_4, x_5 and x_6 belong to class B.

1p **5a** Write down a feature transformation f such that the transformed points $f(x_1)$, $f(x_2)$, . . . , $f(x_6)$ are linearly separable into classes A and B.

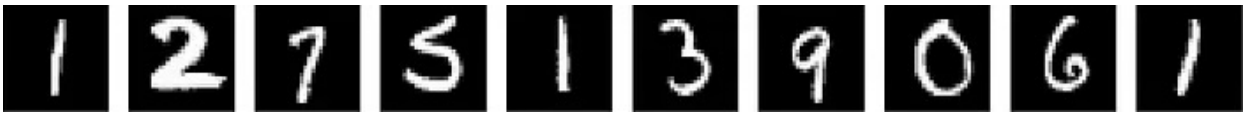
1p **5b** Draw your 6 **transformed** points from the previous question on the one-dimensional line. Then, draw the decision boundary given by the hard-margin linear SVM, and indicate which points are support vectors.

1p **5c** We change the classifier to a soft-margin linear SVM with penalty parameter $C = 3$. What will be the new decision boundary?

1p **5d** What is the kernel $k(x, x')$ associated with the feature function you defined in question a?

[4p] Classification and dimensionality reduction on the MNIST Dataset

MNIST is a standard benchmark dataset consisting of grayscale images of handwritten digits (0–9). Each image has size 28×28 pixels, which can be flattened into a 784-dimensional vector. Each pixel is represented as a scalar value bounded between 0 and 255, indicating the intensity of the grayscale color, where 0 corresponds to black and 255 corresponds to white. For many machine learning models, these values are often rescaled to the range [0, 1] to improve numerical stability and training performance.



Some samples from the MNIST Dataset.

1.5p **6a** We want to use multinomial logistic regression with L1 regularization to classify MNIST images into their $K = 10$ mutually exculsive digit classes.

1. Write the mathematical formulation of the model, including all necessary functions to obtain a predicted class from an input sample.
2. Write down the loss function used to train the model.

*For full points, **you must clearly define all the variables** included in your mathematical expressions.*

1p **6b** Explain how L1 regularization can be leveraged to reduce the dimensionality of the data in this case.

1.5p **6c** We build a standard autoencoder with *one hidden laye*r that reduces the dimensionality of MNIST images to a 64-dimensional latent space.

1. Draw a sketch of the autoencoder, clearly indicating the dimensionality of each layer.
2. How many trainable parameters (including the biases) does the autoencoder have?

[8p] Mixture of Experts for modelling customer behaviour

We want to model the purchasing behavior of customers in a grocery store. The dataset consists of N independent data points, $\mathbf{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$, where each vector $\boldsymbol{x}_n$ has E entries. Each entry x_{ni} represents the quantity of products customer n bought from category i.

Each customer shops according to one of K distinct shopping behavior types, which determine the likelihood of purchasing items from different categories. We introduce a latent variable z to represent the shopping behavior class. We will denote the case $z = k$, i.e., z is the kth class, using a one-hot encoding vector $\mathbf{z} = (z_1, \ldots, z_K) \in \{0, 1\}^K$ with component $z_k = 1$ if $z = k$ and the other components zero. The latent variable has a prior distribution:

$$p(z_k = 1) = \pi_k, \quad \pi_k \in [0, 1], \quad \sum_{k=1}^{K} \pi_k = 1\,.$$

We can then write the full generative model for $\boldsymbol{x}$ (the customer's item purchases) as:

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k p(\boldsymbol{x}|z_k = 1)$$

where each of the $p(\boldsymbol{x}|z_k)$ is a multinomial distribution with T trials (total items bought) and E exclusive events (different item categories):

$$p(\boldsymbol{x}|z = k) = \mathrm{Mult}(\boldsymbol{x}|T, E, \boldsymbol{p}_k) = \frac{T!}{x_1! \ldots x_E!} p_{k1}^{x_1} \ldots p_{kE}^{x_E}\,.$$

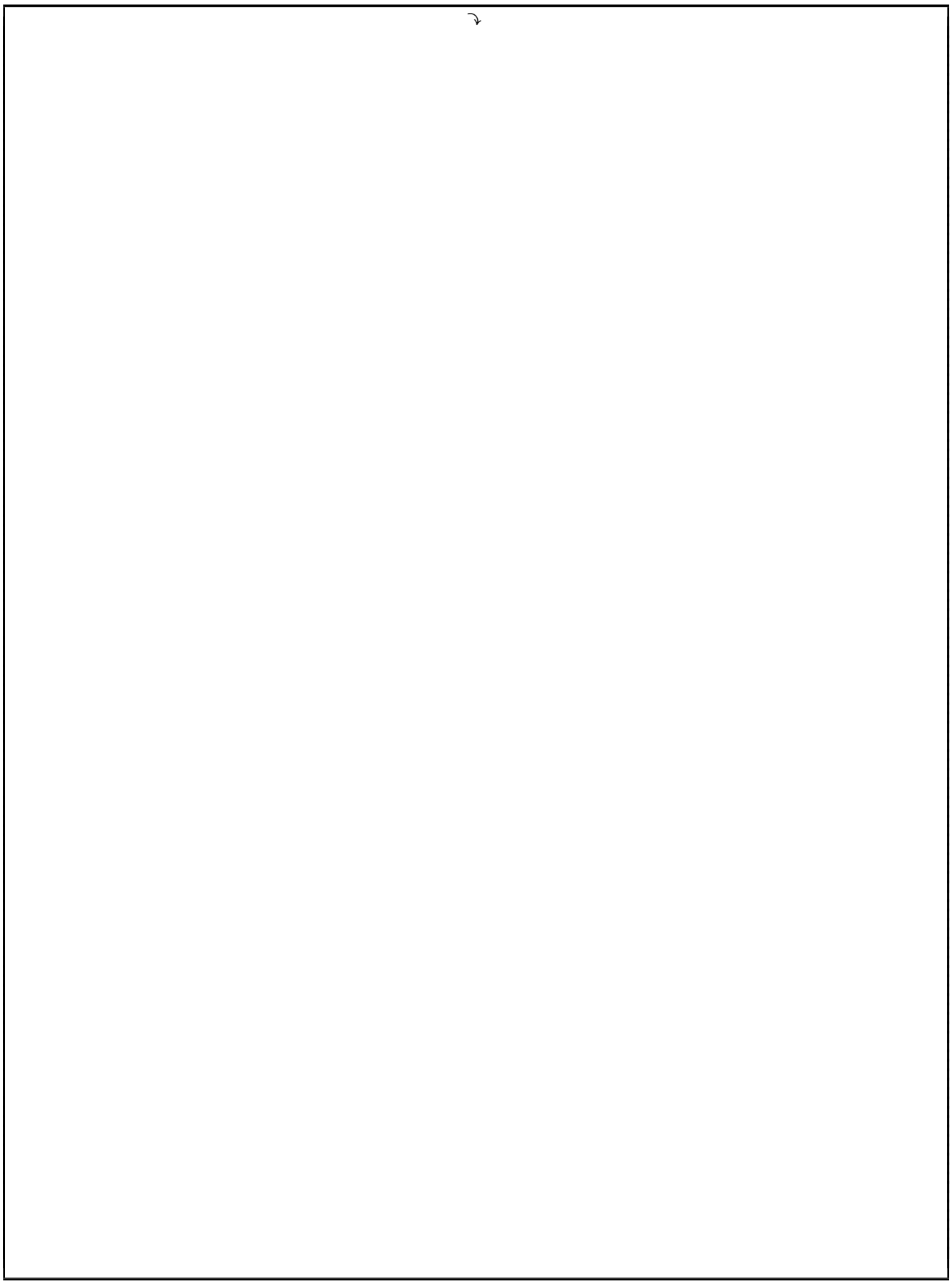
Here, $\boldsymbol{p}_k$ is a vector of probabilities for the item categories: $0 \leq p_{ki} \leq 1$ for $i = 1, \ldots, E$. For each customer category, the probabilities of the items sum to one: $\sum_{i=1}^{E} p_{ki} = 1$.

1p **7a** Derive the expression for the responsibilities $r_{nk} = p(z_k = 1|\boldsymbol{x}_n)$. If you prefer, you don't have to explicitly write out the expression for Mult.

1p **7b** Write down the log-likelihood $(\ln p(\boldsymbol{X}))$ for the dataset. Write out the distributions explicitly.

3p **7c** Give the update rule that maximizes likelihood for the parameter p_{ki}. Remember to account for the constraint $\sum_{i=1}^{E} p_{ki} = 1$. Make sure the update rule is expressed in terms of the responsibilities r_{nk}, in order to get a tractable expression.

↳



1.5p **7d** Suppose you have derived the maximum likelihood solution for π_k too, just like you would have obtained in the previous exercise for p_{ki}. Formulate the EM algorithm the maximizes the log-likelihood w.r.t. parameters $\{\boldsymbol{p}_k\}_{k=1}^K$ and $\{\pi_k\}_{k=1}^K$. Make sure to also provide a suitable stopping criterion.

1.5p **7e** Regarding the EM algorithm for mixture models:
a) Is the algorithm guaranteed to converge?
b) Will it always find the global maximum of the log-likelihood?
c) What can we say about how the log-likelihood changes between iterations?"

Emergency box

8 Use this space only as a last resort to provide answer to a previous question that didn't fit in the corresponding box. Indicate which question it is about.